

CLIMATE CHANGE AND WEST AFRICA'S AQUATIC ECOSYSTEMS

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What is Climate Change?

Climate change refers to long-term changes in the Earth's climate caused primarily by human activities, especially the burning of fossil fuels, deforestation, and land-use changes that increase greenhouse gases such as carbon dioxide (CO₂) and methane (CH₄) in the atmosphere. These gases trap heat, causing global temperatures to rise and disrupting natural climate systems.



Impacts of Climate Change

- Rising air and ocean temperatures
- Sea-level rise
- More intense rainfall and flooding
- Longer droughts and heatwaves
- Stronger coastal storms
- Changes in ocean circulation and weather patterns

How Climate Change is Affecting Aquatic Ecosystems in West Africa

- ❑ *Declining Fisheries and Food Security* - Warmer waters, changing ocean currents, and habitat degradation are altering fish migration, breeding, and survival. Many commercially important fish stocks are shifting distribution or declining, threatening food security and livelihoods.
- ❑ *Loss of Mangroves and Wetlands* - Sea-level rise, coastal erosion, and human pressures reduce the health of mangrove forests and wetlands that protect shorelines, store carbon, filter water, and provide breeding grounds for fish and shellfish.
- ❑ *Coastal Erosion and Flooding* - Many West African coastal communities experience increasing shoreline erosion and flooding due to rising sea levels, stronger storm surges, and the loss of natural coastal buffers.
- ❑ *Declining Water Quality* - Heavy rainfall and flooding wash sediments, nutrients, plastics, and pollutants into rivers and coastal waters, reducing water quality and damaging aquatic ecosystems.

Advancing Climate Resilience

Climate change threatens West Africa's waters, but healthy ecosystems and informed communities can be powerful allies in building a resilient future. WASMaN is committed to advancing climate resilience through science, policy advocacy, community engagement, capacity building, and partnerships that strengthen the sustainable management of West Africa's aquatic ecosystems. We champion inclusive solutions that recognize the vital role of women in protecting aquatic resources and building climate-resilient communities.